

CS 151: Operating Systems Fundamentals and Security

October 14th, 2025

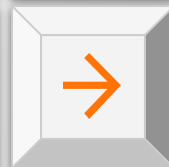
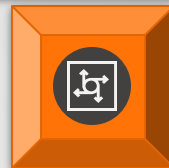
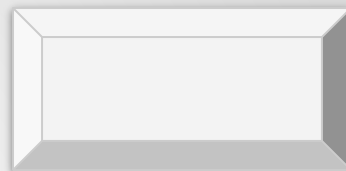


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01

OS Basics



What is an **Operating System**?



A piece of software that manages **computer hardware** and **software resources**.

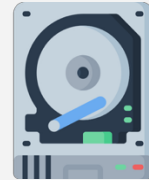
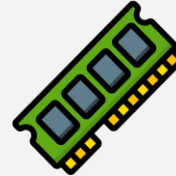
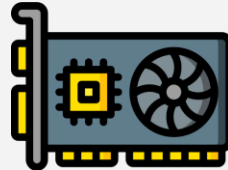
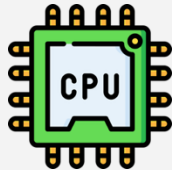
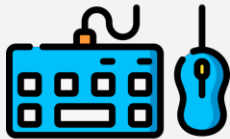
Responsible for:

- Process Scheduling
- Memory Management
- I/O (Input-Output) Management
- File Management
- User Interface

What is an **Operating System**?



Operating System



Operating System Key Terms & Concepts

Kernel – Core part of the OS, responsible for acting as a “bridge” between software and hardware resources.

Shell – Program that provides an interface to the kernel.

- Why shell? Outer layer of the kernel

Operating System Key Terms & Concepts

32-bit vs 64-bit

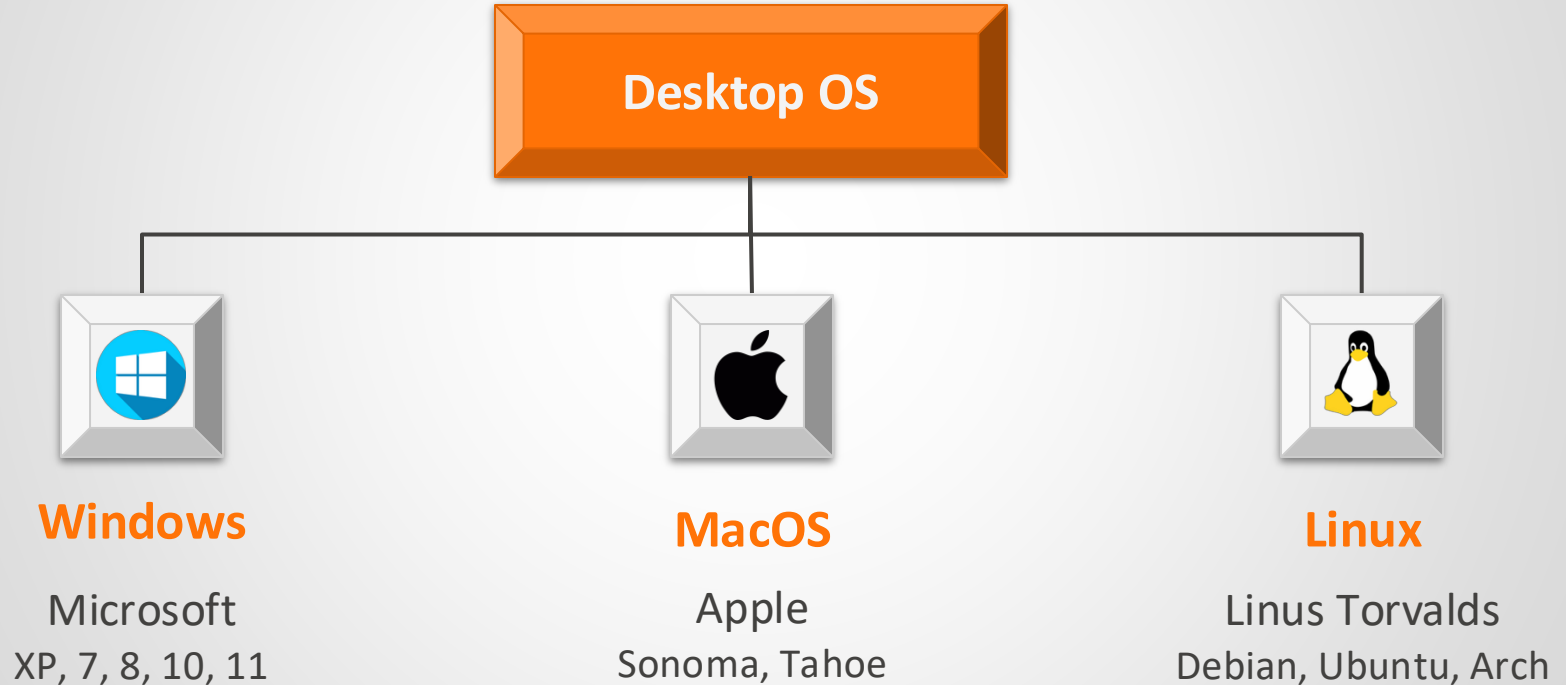
- Refers to the type of **processor** that can be supported
- **32-bit (x86)** vs. **64-bit (x64)**
 - $2^{32} = 4,294,967,296$ values
 - $2^{64} = 18,446,744,073,709,551,616$ values
- 32-bit OS **CANNOT** run 64-bit drivers/applications, but 64-bit OS can run 32-bit applications

02

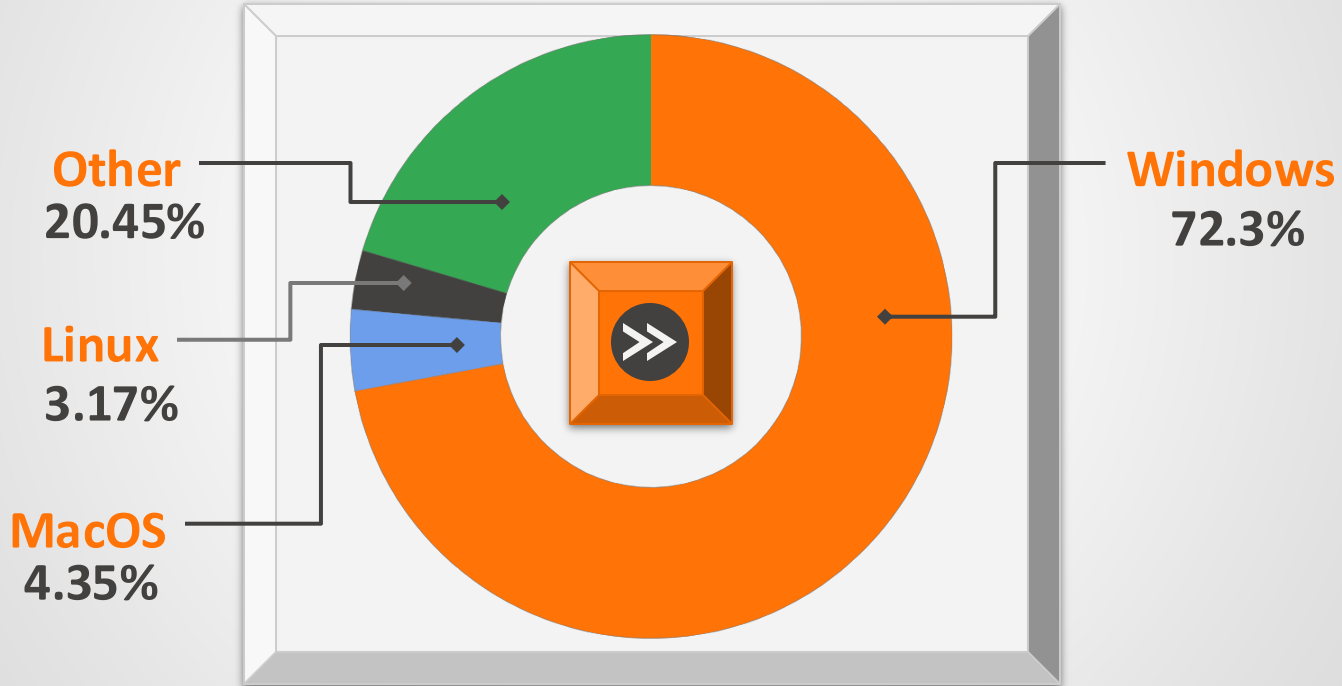
Types of OS



Desktop **Operating Systems** Families

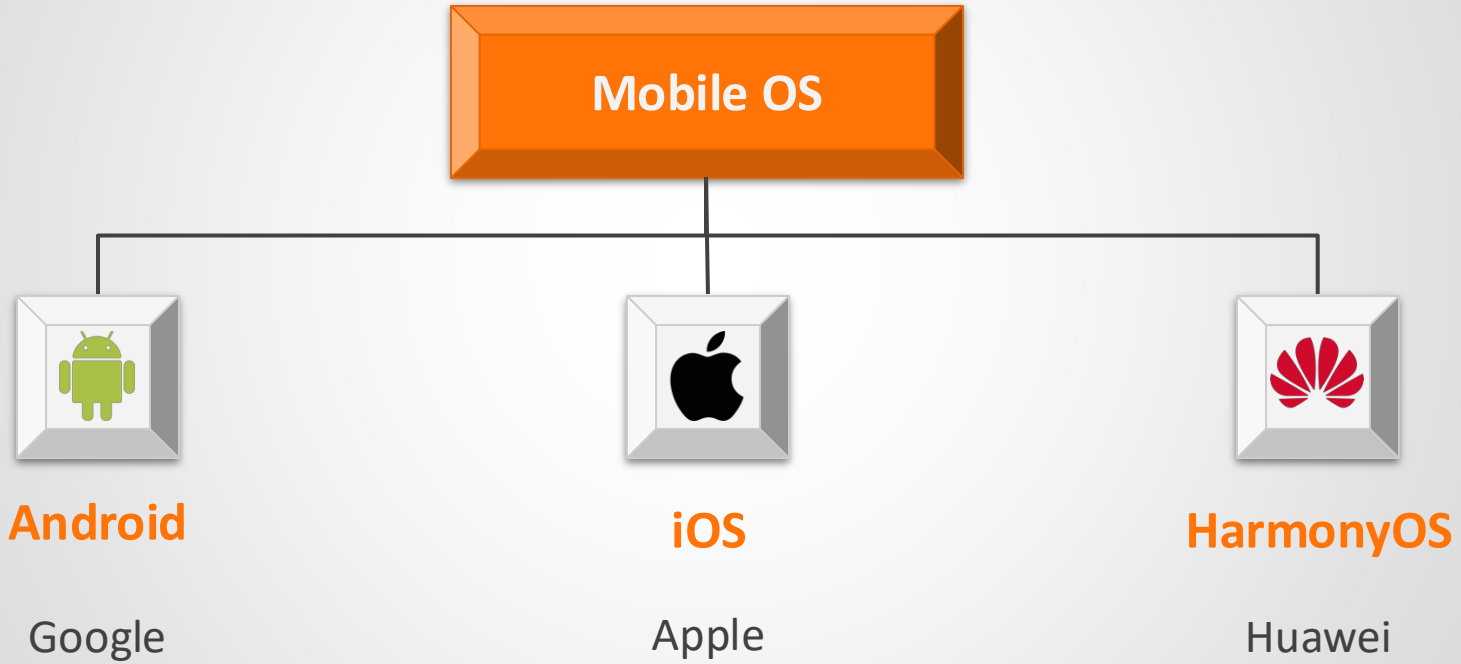


Desktop OS Market Shares

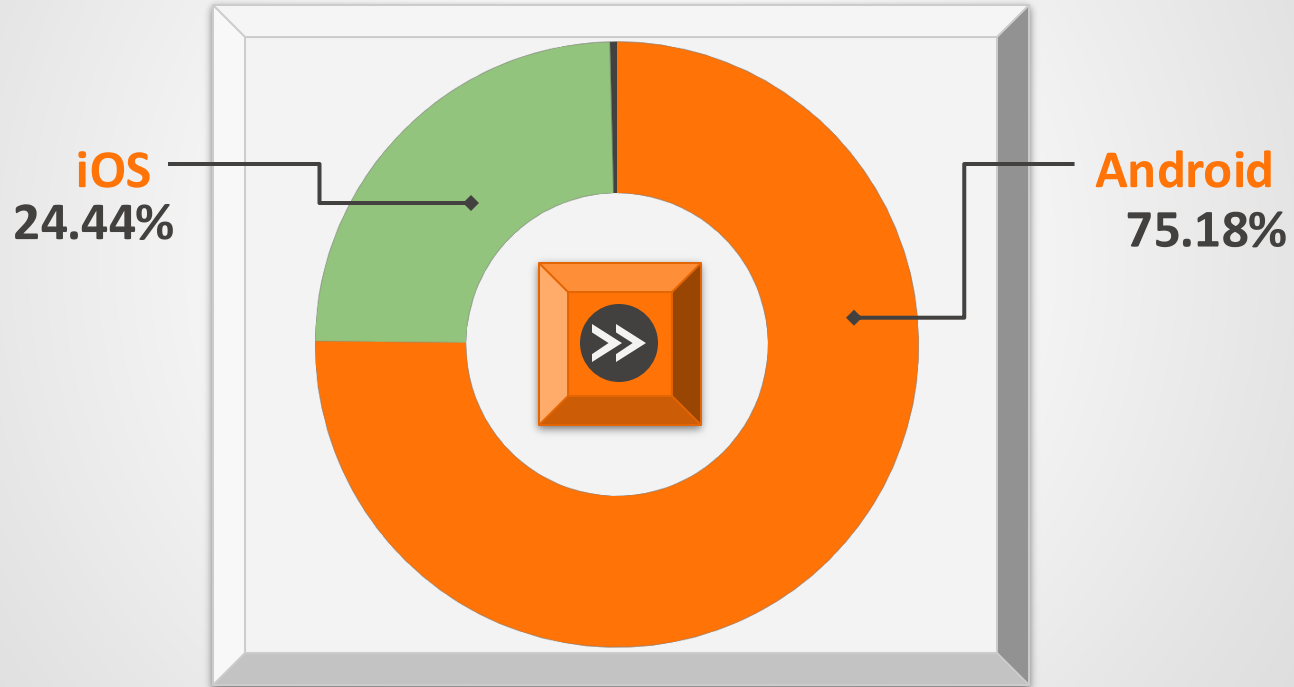


Data from [GlobalStats](#)

Mobile **Operating Systems** Families



Mobile OS Market Shares



Data from [GlobalStats](#)

Windows Operating Systems



Developed by Microsoft in 1985

- Built to provide a GUI for MS-DOS
- Windows NT – Windows New Technologies

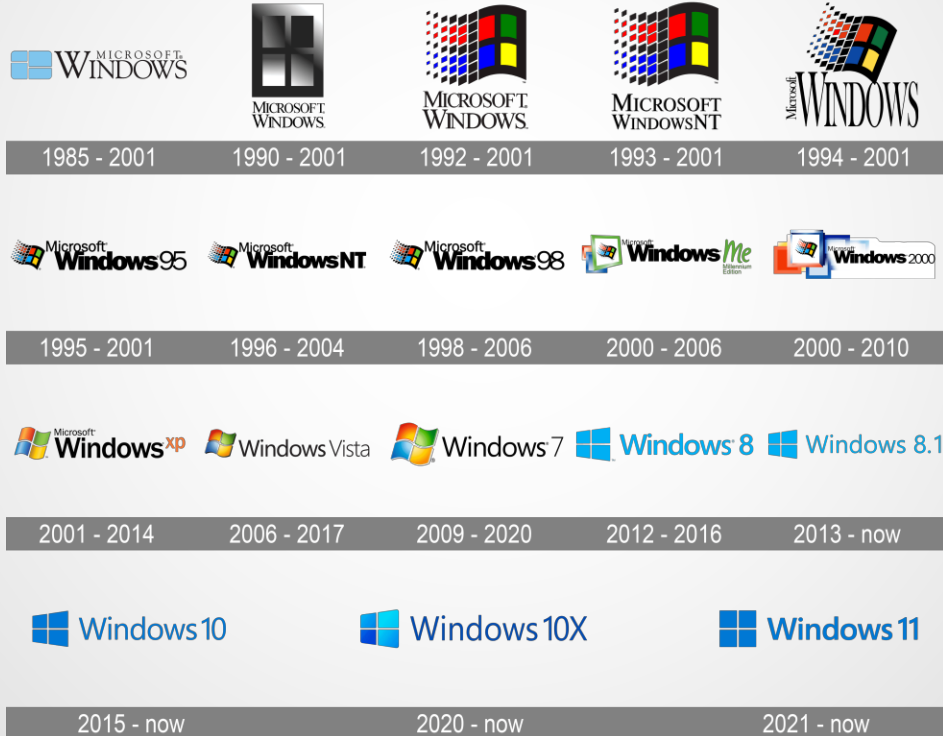
Various editions

- Home, Home Pro, Pro, Education, Enterprise, etc.

Windows Operating Systems



Windows Operating Systems



Mac Operating System



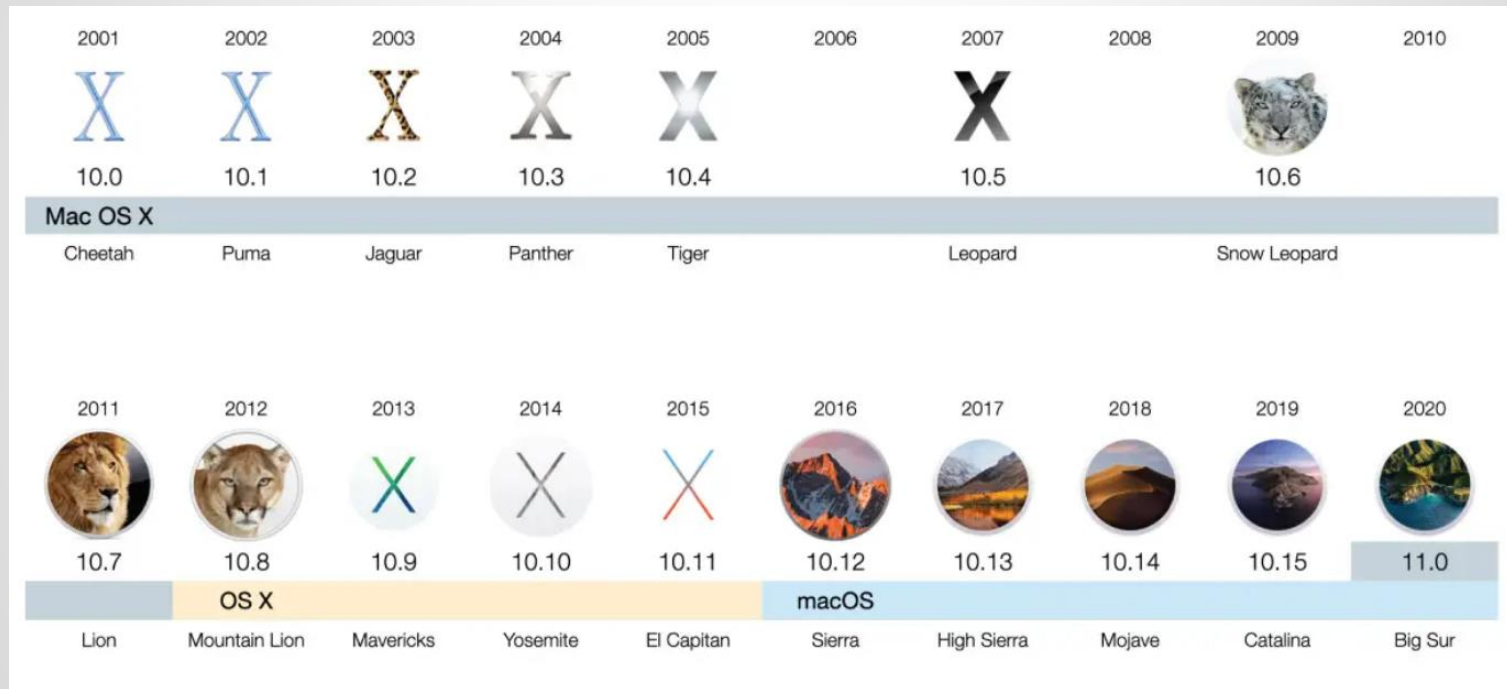
Developed by Apple in 2001

- Unix-like OS
- Derived from OPENSTEP and FreeBSD
- Succeeds classic Mac OS
 - Used by the Macintosh Systems from 1984 - 2001

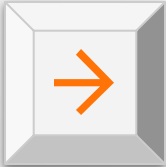
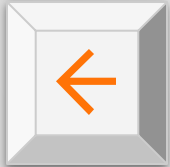
Mac Operating System



Mac Operating System



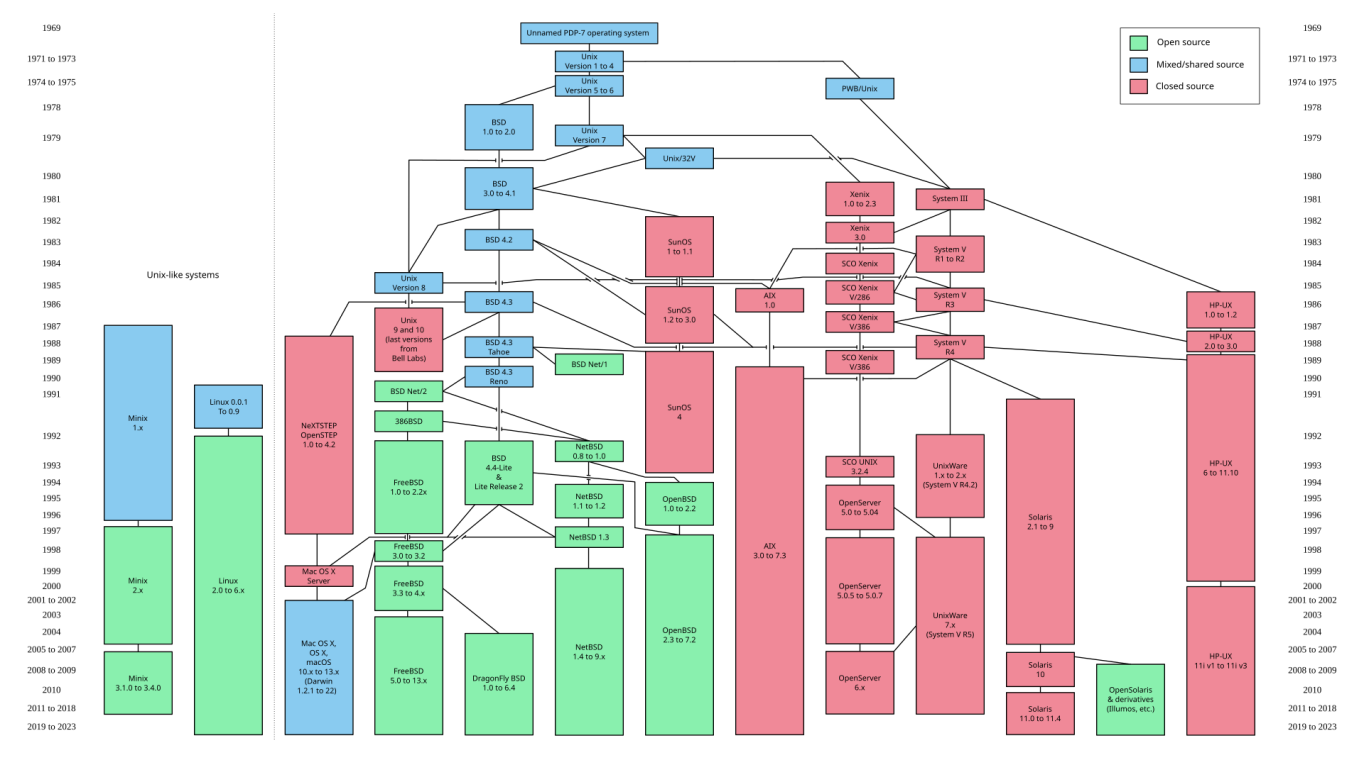
Linux Operating Systems



UNIX is NOT LINUX

- Unix is proprietary, but other OS can be described as Unix-like
- Unix was developed by Ken Thompson and Dennie Ritchie in the 1960s
- Default Shell is the Bourne Shell

Unix Evolution



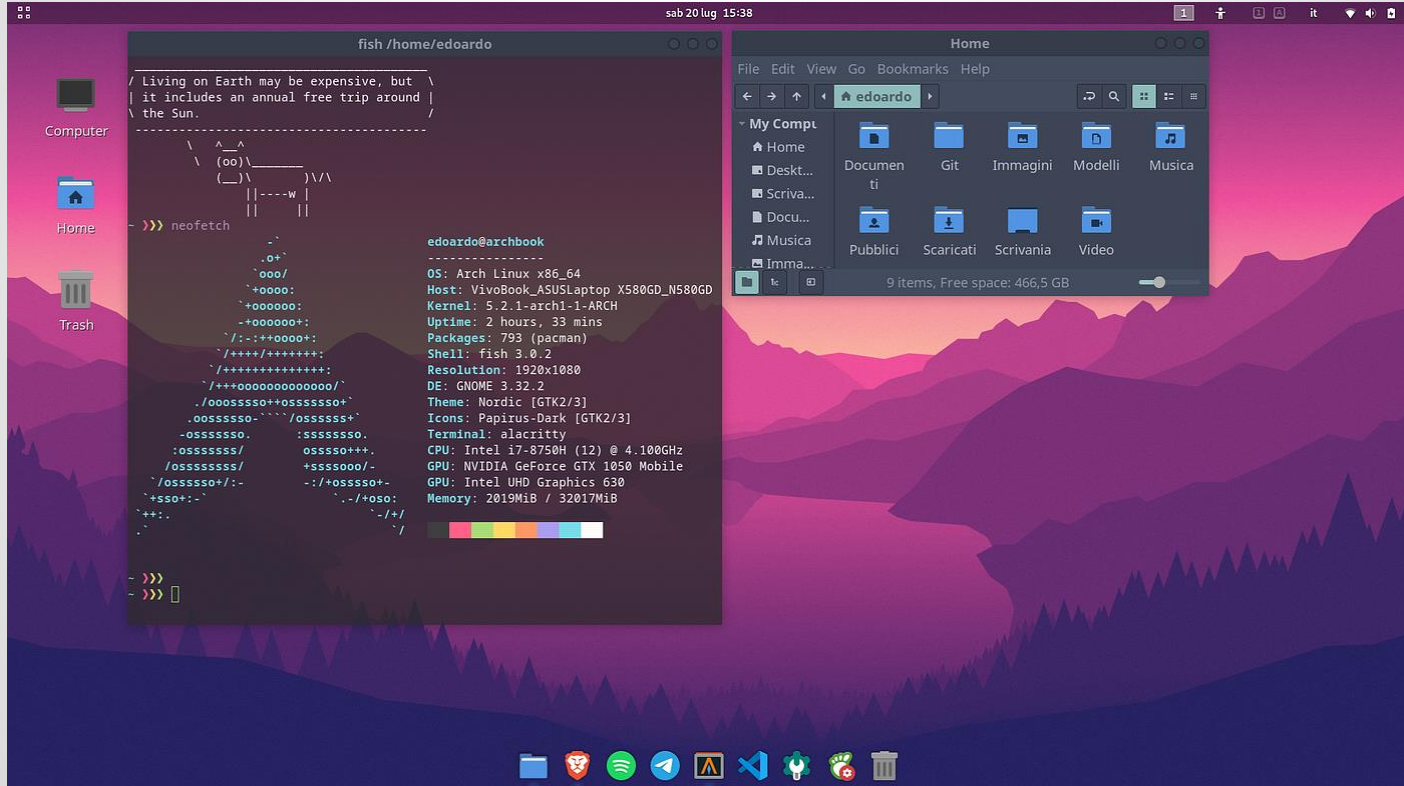
Linux Operating Systems



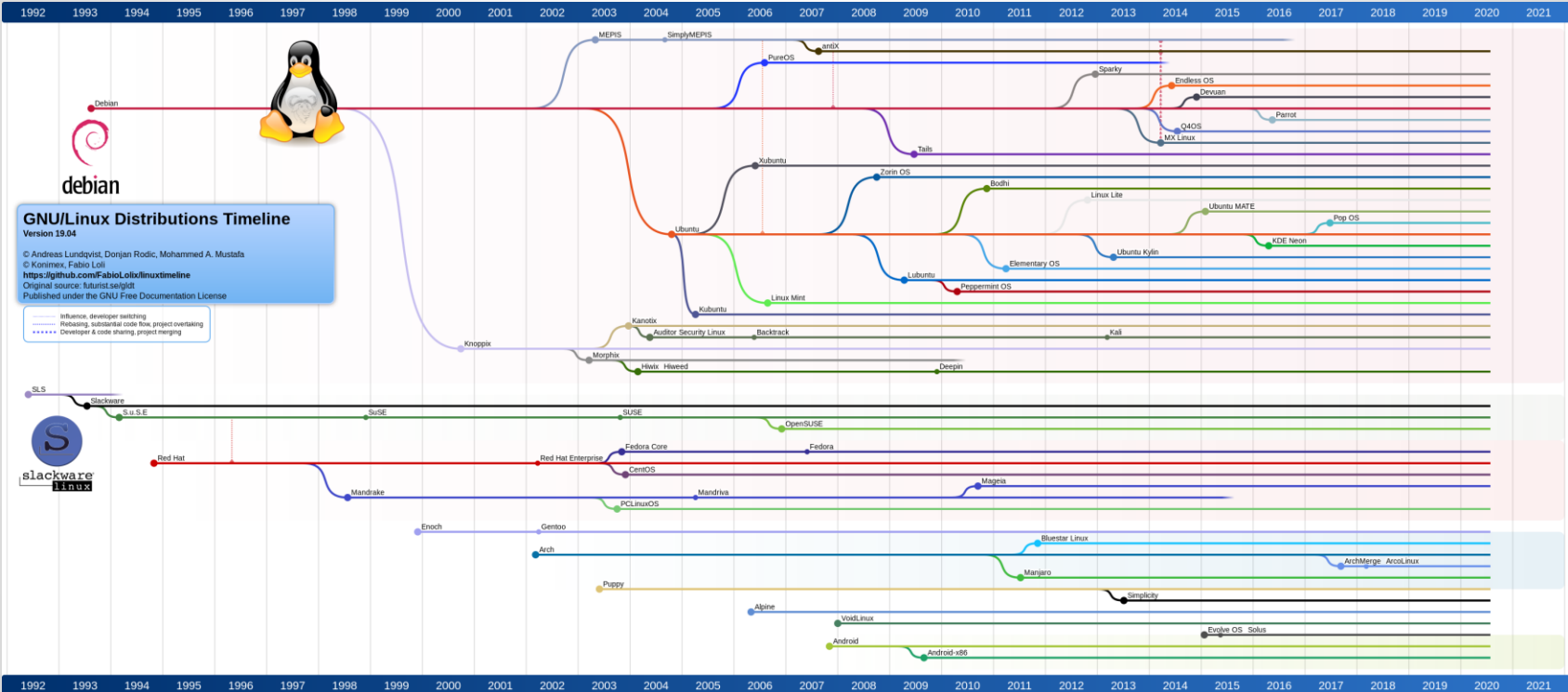
Linux was developed by Linux Torvalds in the 1990s and is typically characterized as free and open-source.

- Ubuntu, Debian, Solaris, RedHat, Arch, etc.
- Default shell is the Bourne Again Shell (BASH)

Linux Operating Systems



Linux Operating Systems



Why do we care?

Security-perspective

- Need be aware of different types of OS, the underlying differences, and limitations between all.
- Need to be able to communicate to other professionals, as well as average users.
- Hardening/Defense begins with the OS, and understanding attack vectors requires OS knowledge

Why do we care about Linux?

- Greater support for open-source/customizable tooling, especially in the networking/cybersecurity space.
- Many Linux Distributions that are *not* general-purpose OS and instead have a specific-purpose and come preinstalled with useful software.
 - Kali Linux, ParrotOS, BlackArch
 - Security Onion, SIFT Workstations (Forensics), REMnux (Reverse Engineering)

03

OS Security



Key Terms & Concepts



Principle of Least Privilege

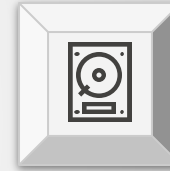
Give only the absolute minimum permissions necessary to perform a required task/function



Defense in Depth

Multiple layers of security controls should be implemented

(Physical, Technical, Administrative)



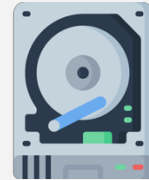
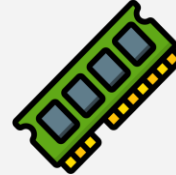
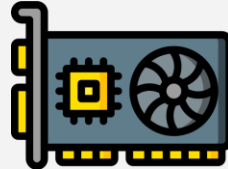
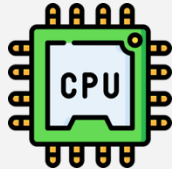
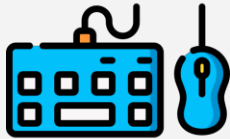
Open-Design

Security should *not* depend on secrecy of its design.

User-Space vs Kernel-Space



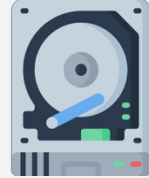
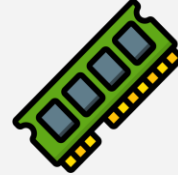
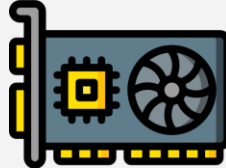
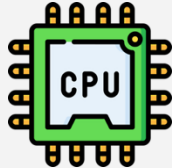
Operating System



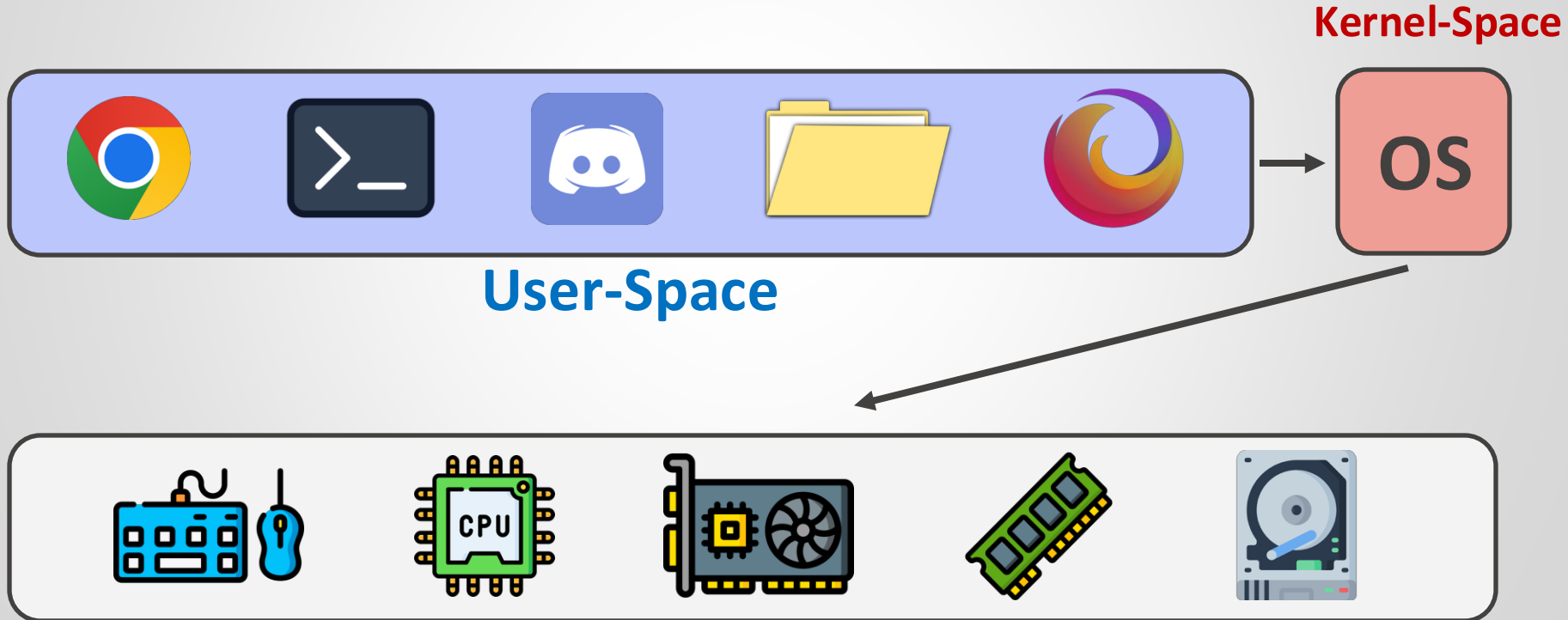
User-Space vs Kernel-Space



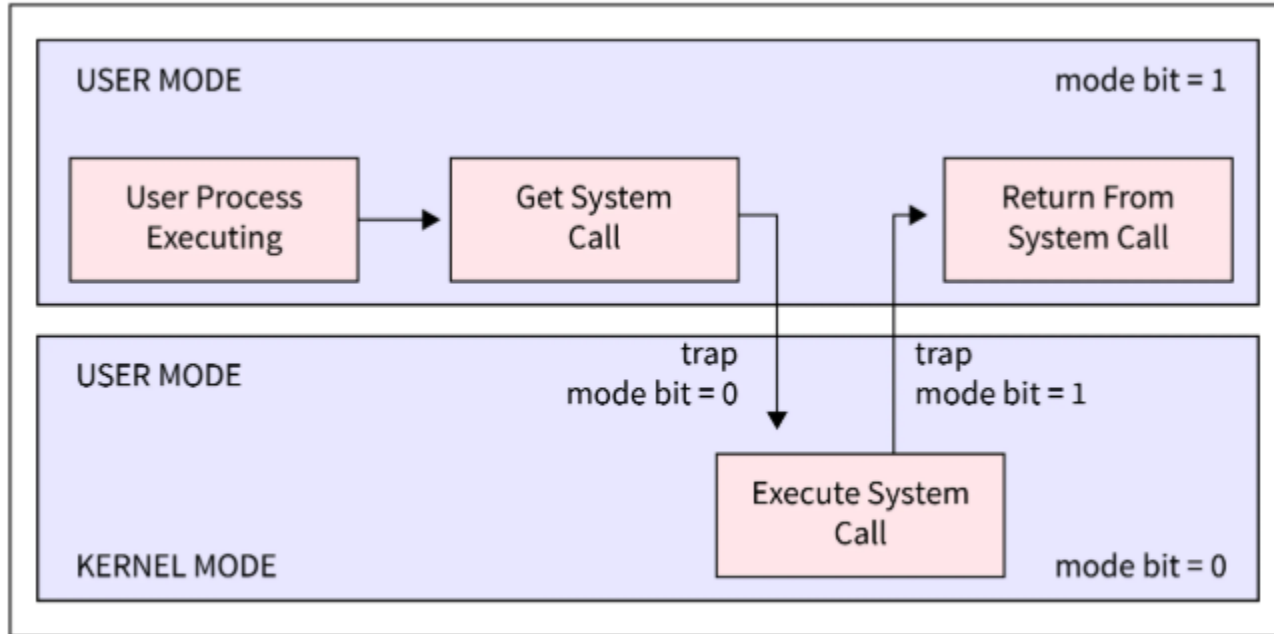
OS



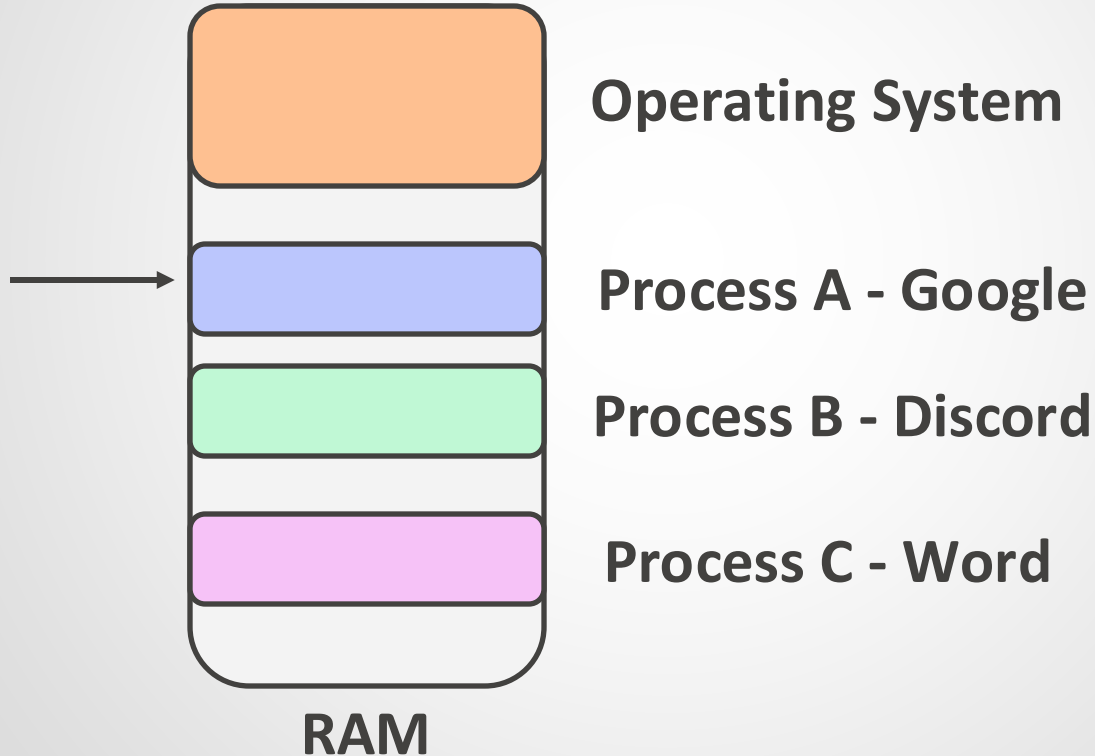
User-Space vs Kernel-Space



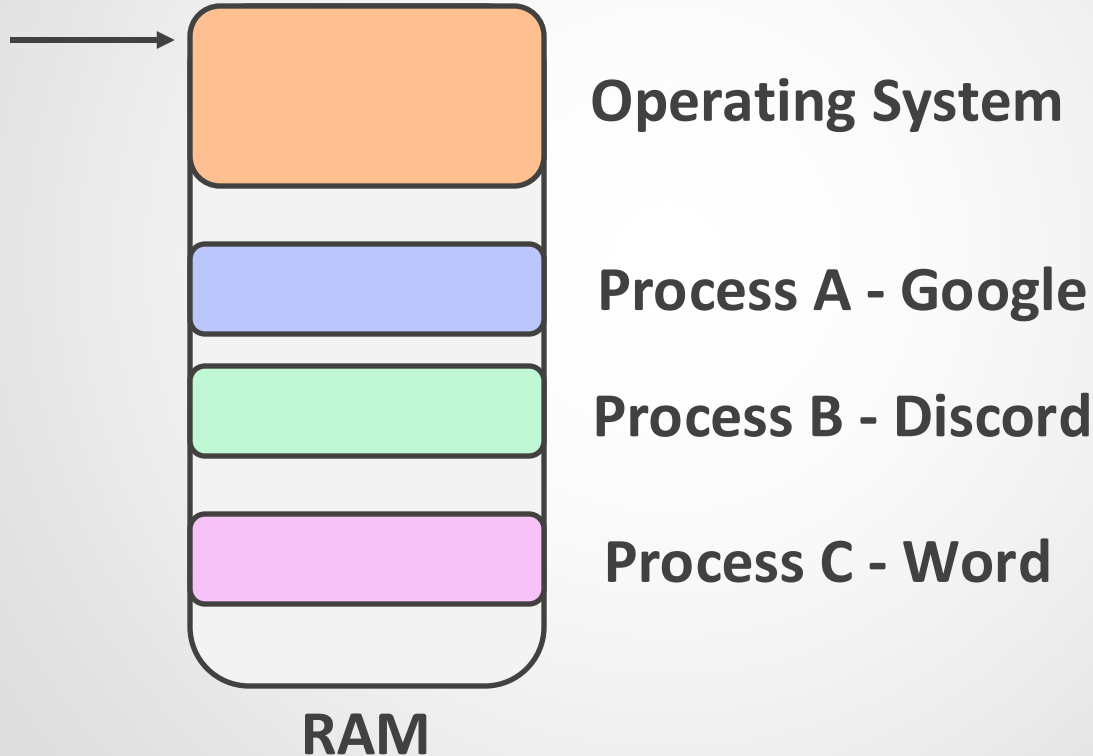
User-Space vs Kernel-Space



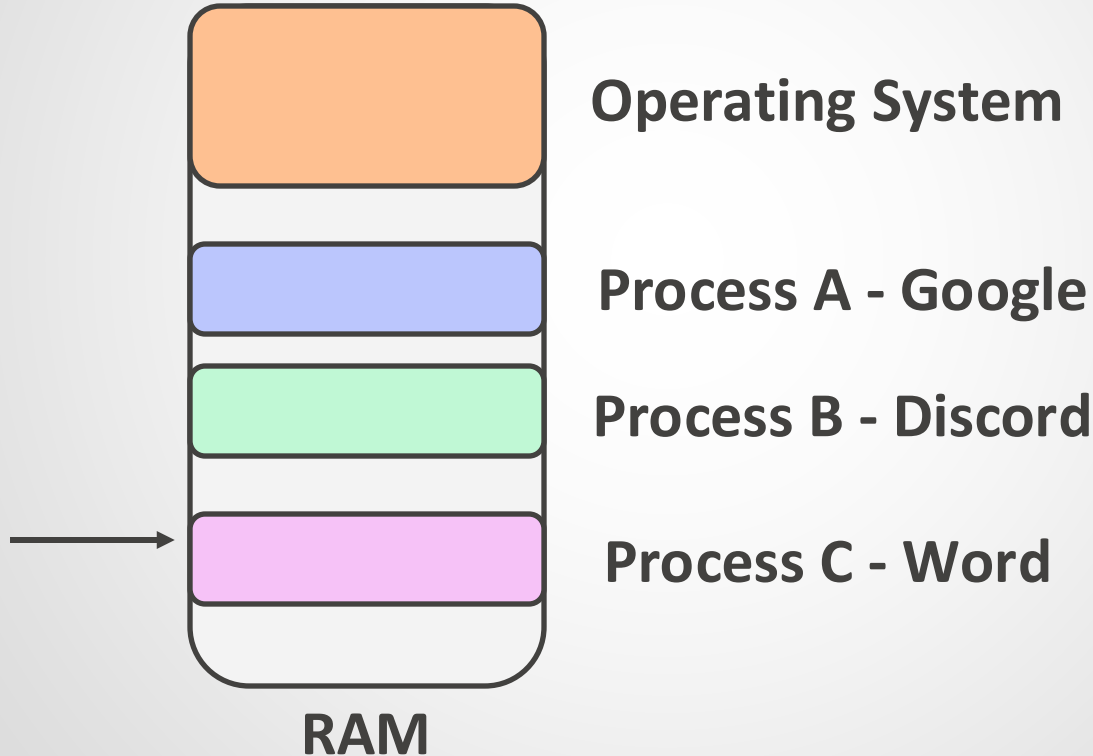
User-Space vs Kernel-Space



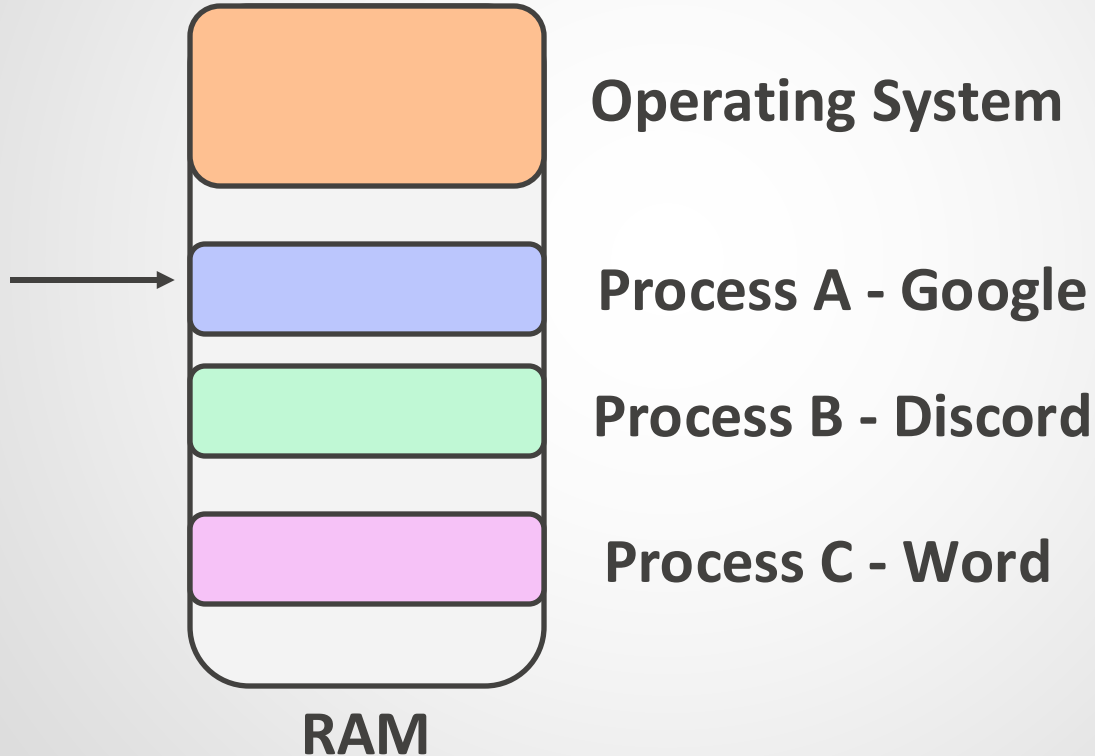
User-Space vs Kernel-Space



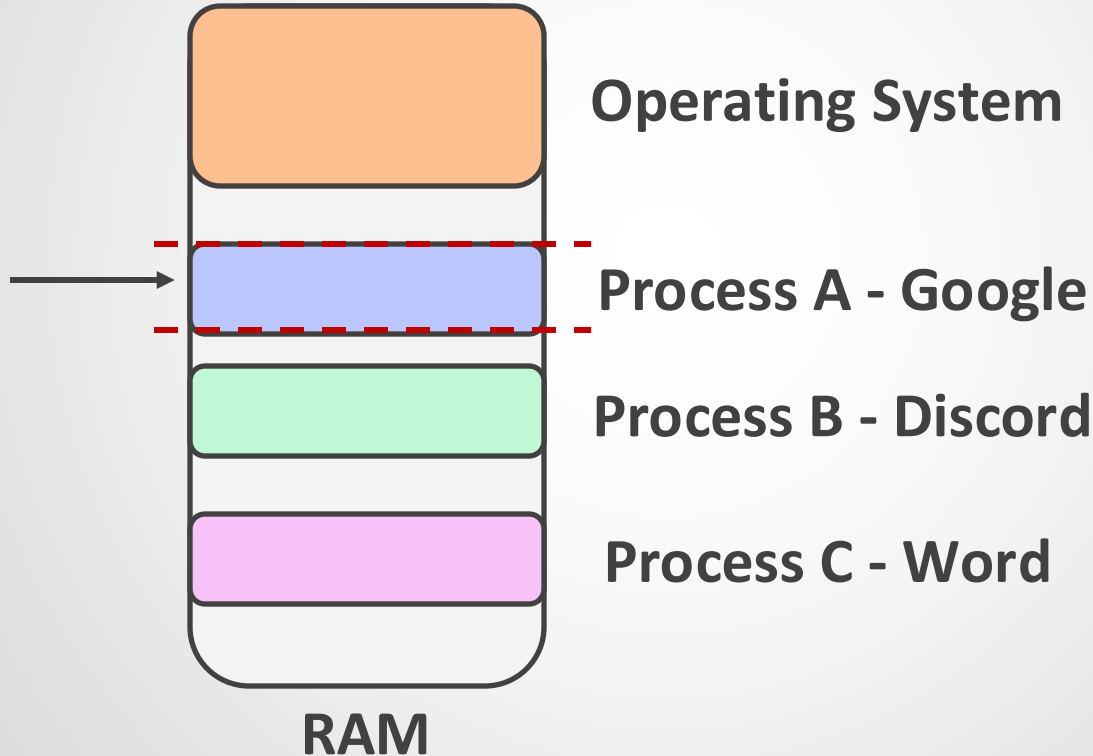
User-Space vs Kernel-Space



User-Space vs Kernel-Space

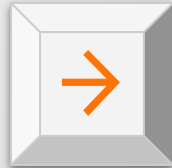
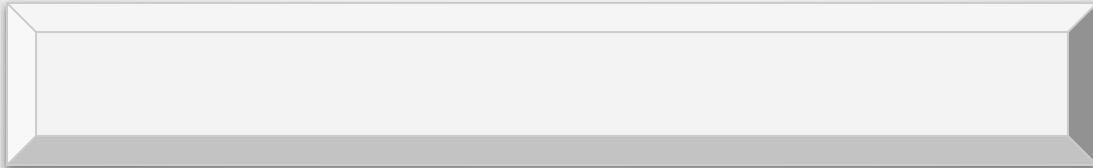
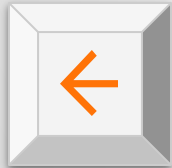


User-Space vs Kernel-Space





Access Control



Access Control Key Terms & Concepts

Subjects – Model for users, application, etc. Wants to access a resource/object.

Objects – Data to be interacted with or accessed

Access Control Key Terms & Concepts

Access Control Lists – Also known as ACLs, a list that describes who can access what (files, network resources, etc.)

Authorization – Granting (or denying) permission to perform an action

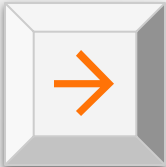
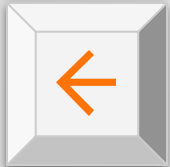
Authentication – Determining a subject is who they say they are

Access Control Key Terms & Concepts

Users – Individual within a computing system.

Groups – Set of people within a single unit.

Access Control Models



There are various different types of Access Control Models:

- Discretionary
- Mandatory
- Role-Based

Discretionary Access Control (DAC)

Owner of an object controls who can read, write, or execute the given resource.

Owner can give access to users or groups.

Mandatory Access Control (MAC)

Access to resources is determined by predefined security-levels.

Object-focused, not user-focused.

Mandatory Access Control (MAC)

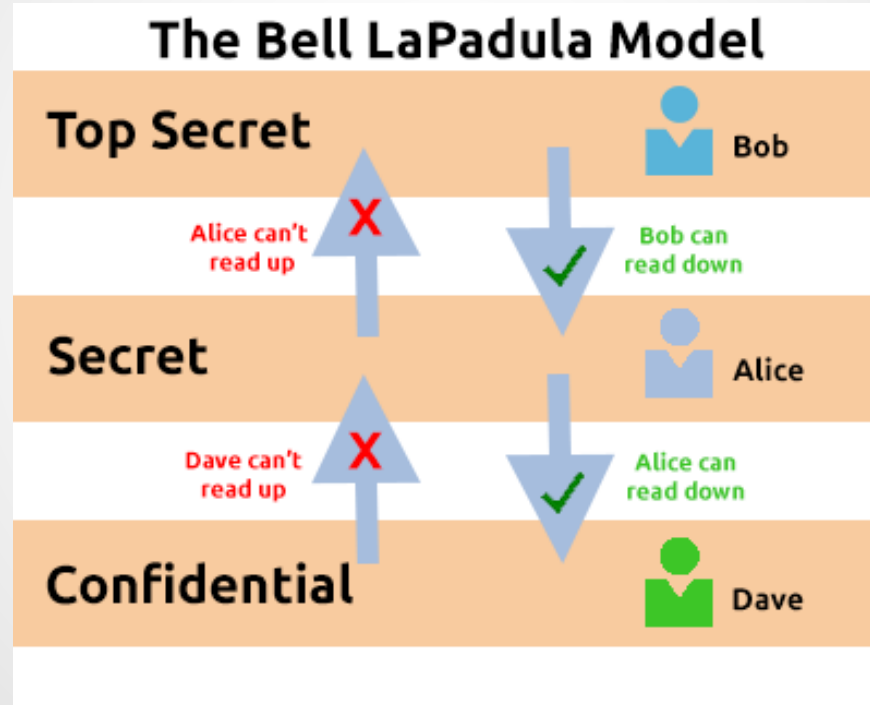
Military Classification

TOP SECRET
SECRET
CONFIDENTIAL
UNCLASSIFIED

Business Classification

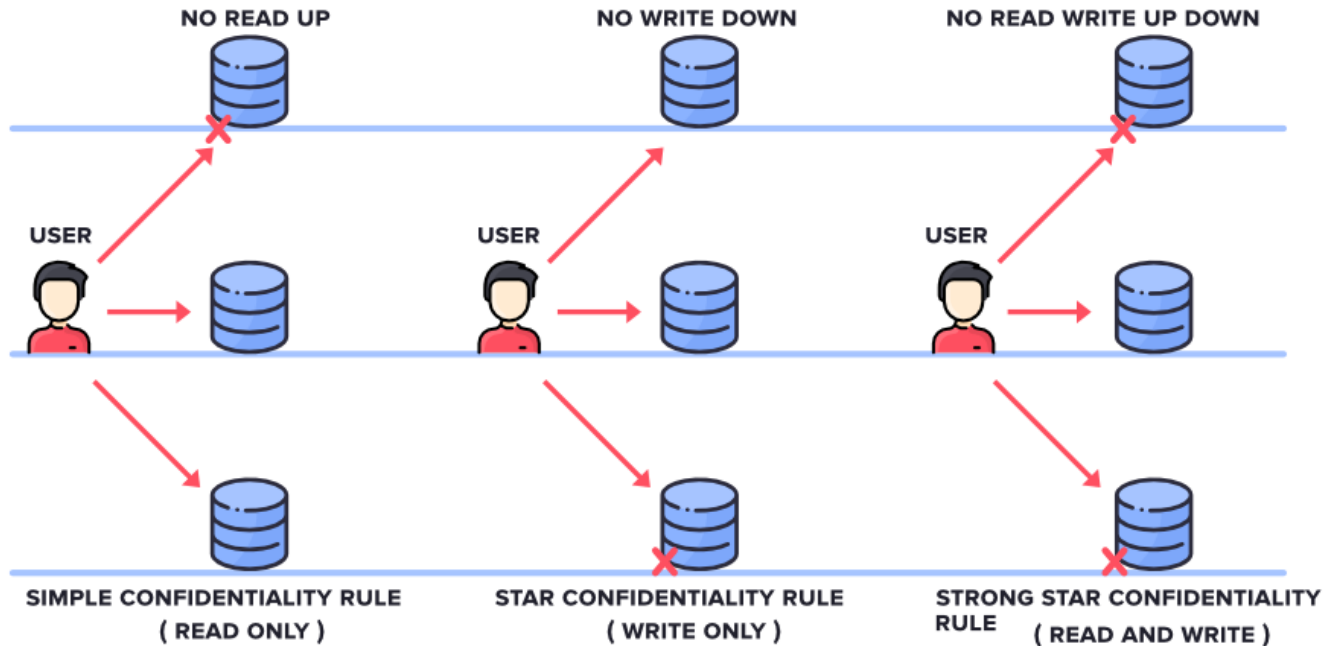
HIGHLY SENSITIVE
SENSITIVE
INTERNAL
PUBLIC

Bell – Lapadula Model



Bell – Lapadula Model

BELL - LAPADULA MODEL

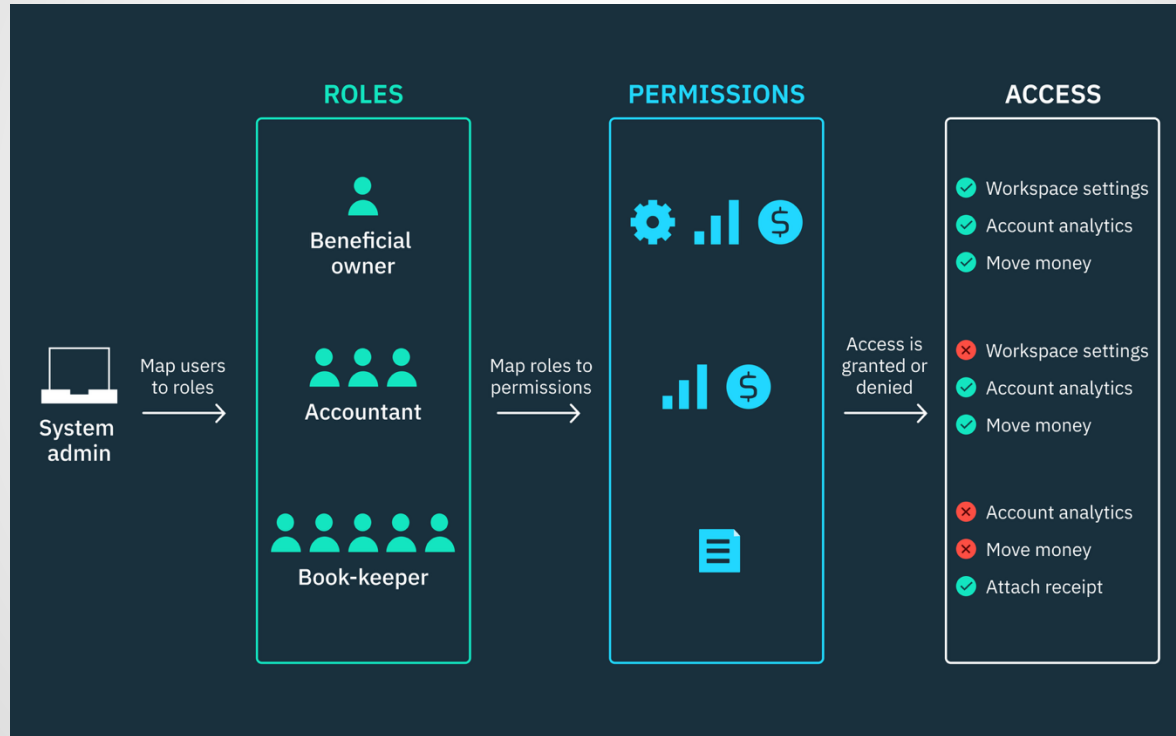


Role-Based Access Control (RBAC)

System Administrator creates roles, in which users or groups within a role obtain the corresponding permissions.

User-focused, not object-focused.

Role-Based Access Control (RBAC)



Thank **you!**

Do you have any questions?



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